

#This code will be used to demonstrate lemma 6.6. We demonstrate the non —existence of the curve γ in region A2 and A3.

#Construct the polynomial R from Theorem 6.1.3.

$$A := -(w + z)^3 ((x^2 y^2 + z^4) w^4 - x^2 y^2 z^4) :$$

$$B := x^2 y^5 w^2 z^2 (w + z)^2 (w - z) - x^2 y^2 (w + z)^3 (w^2 + z^2) w^2 z^2 (w - z) + w^4 y^3 z^4 (w + z)^3 + w^4 z^4 x^3 y^3 :$$

$$R := B^2 \cdot ((w + z)^2 - y^2)^3 - A^2 \cdot (y^2 \cdot z \cdot w)^3$$

$$R := (x^2 y^5 w^2 z^2 (w + z)^2 (w - z) - x^2 y^2 (w + z)^3 (w^2 + z^2) w^2 z^2 (w - z) + w^4 y^3 z^4 (w + z)^3 + w^4 z^4 x^3 y^3)^2 ((w + z)^2 - y^2)^3 - (w + z)^6 ((x^2 y^2 + z^4) w^4 - x^2 y^2 z^4)^2 y^6 z^3 w^3 \quad (1)$$

#Apply the rational parametrization constructed in this chapter 4..

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify(numer(expand(subs(parametrization, R)))) ;

$$\begin{aligned} & -(a + b)^{12} \left(-4 b^2 (b^{28} + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} \right. \\ & \quad + 78414 b^{14} + 92394 b^{12} + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} \\ & \quad + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} \\ & \quad - 12876 b^8 - 348 b^6 + 3252 b^4 + 420 b^2 - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} \right. \\ & \quad + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 + 2369 b^6 - 699 b^4 + b^2 + 1 \Big) b^6 a^{22} - 9 \left(b^{16} + 8 b^{14} \right. \\ & \quad - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 - 528 b^4 + \frac{4168}{9} b^2 + 1 \Big) b^8 a^{24} - 1024 b^{15} (b^2 \\ & \quad + 1)^4 a^7 + 2 b^4 (b^{24} + 36 b^{22} - 2636 b^{20} - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} \end{aligned} \quad (2)$$

$$\begin{aligned}
& - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} \\
& + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} \\
& + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} - 1536 (b^2 + 1)^4 b^9 \left(b^8 - b^6 + \frac{5}{3} b^4 \right. \\
& \left. - \frac{16}{3} b^2 + \frac{5}{3} \right) a^{13} - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 - 1 \right) (b^2 + 1)^4 b^{11} a^{11} - 1024 \left(b^2 \right. \\
& \left. - \frac{1}{2} \right) (b^2 + 1)^5 b^{13} a^9 + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} \\
& + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} + 5632 (b^2 + 1)^4 b^7 \left(b^6 \right. \\
& \left. - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) a^{21} - 3584 \left(b^8 - \frac{52}{7} b^6 + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \right) (b^2 \\
& + 1)^4 b^7 a^{19} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} \\
& + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} \\
& - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\
& - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} \\
& - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\
& + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\
& + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\
& + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\
& - 4 b^{14}) a^2 + a^{32} b^{16} + b^{16} - 2560 (b^2 - 2) \left(b^2 - \frac{1}{5} \right) (b^2 + 1)^4 b^7 a^{23} \Big) (b^2 \\
& + 1)^3 (a b - 1)^6 (a^2 + 1)^3 (a b + 1)^6
\end{aligned}$$

#Define r1

$$\begin{aligned}
r1 := & \left(b^{16} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \right. \\
& \left. + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} + 2 b^4 (b^{24} \right.
\end{aligned}$$

$$\begin{aligned}
& + 36 b^{22} - 2636 b^{20} - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 \\
& - 1372 b^6 + 180 b^4 + 36 b^2 + 1) a^{12} - 1024 b^{15} (b^2 + 1)^4 a^7 - 4 b^2 (b^{28} + 12 b^{26} + 37 b^{24} \\
& - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} + 36360 b^{10} + 5211 b^8 \\
& - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 2396 b^{18} - 25164 b^{16} \\
& - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 + 420 b^2 - 127) a^{20} \\
& + 8 b^6 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 + 2369 b^6 - 699 b^4 \right. \\
& \left. + b^2 + 1 \right) a^{22} - 9 b^8 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 - 528 b^4 + \frac{4168}{9} b^2 \right. \\
& \left. + 1 \right) a^{24} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} \\
& + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} \\
& - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\
& - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} \\
& - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\
& + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\
& + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\
& + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\
& - 4 b^{14}) a^2 + a^{32} b^{16} - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 + 1)^4 (b^2 - 2) a^{23} - 1536 \left(b^8 - b^6 + \frac{5}{3} b^4 \right. \\
& \left. - \frac{16}{3} b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 - 1 \right) b^{11} (b^2 + 1)^4 a^{11} \\
& - 1024 b^{13} (b^2 + 1)^5 \left(b^2 - \frac{1}{2} \right) a^9 + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 \\
& + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} \\
& + 5632 b^7 (b^2 + 1)^4 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) a^{21} - 3584 \left(b^8 - \frac{52}{7} b^6 + \frac{95}{7} b^4 \right. \\
& \left. - \frac{43}{7} b^2 + \frac{3}{7} \right) b^7 (b^2 + 1)^4 a^{19} \Big):
\end{aligned}$$

collect(expand(r1), a)

$$(-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + (72 b^{20} - 76 b^{18} \quad \mathbf{(3)}$$

$$\begin{aligned}
& + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} - 120 b^{26} - 3426 b^{24} \\
& - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} - 157780 b^{12} - 32376 b^{10} \\
& + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} - 8440 b^{24} - 21976 b^{22} \\
& - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} + 8 b^8 + 8 b^6) a^{10} + (\\
& - 265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} + 144 b^{12} - 72 b^{10} - 9 b^8) \\
& a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} + 72 b^{12}) a^6 + (-250 b^{20} \\
& - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + a^{32} b^{16} + b^{16} + (\\
& - 9 b^{24} - 72 b^{22} + 144 b^{20} + 648 b^{18} - 3698 b^{16} + 2952 b^{14} + 4752 b^{12} - 4168 b^{10} - 9 b^8) a^{24} \\
& + (-2560 b^{19} - 4608 b^{17} + 6144 b^{15} + 19456 b^{13} + 13824 b^{11} + 1536 b^9 - 1024 b^7) a^{23} \\
& + (8 b^{26} + 8 b^{24} - 216 b^{22} - 1272 b^{20} + 8652 b^{18} - 14616 b^{16} - 4148 b^{14} + 18952 b^{12} \\
& - 5592 b^{10} + 8 b^8 + 8 b^6) a^{22} + (5632 b^{21} + 1024 b^{19} - 32768 b^{17} - 31232 b^{15} + 26112 b^{13} \\
& + 40960 b^{11} + 9216 b^9 - 2560 b^7) a^{21} + (2 b^{28} + 72 b^{26} + 360 b^{24} - 4792 b^{22} - 50328 b^{20} \\
& - 85104 b^{18} - 181788 b^{16} - 169584 b^{14} - 25752 b^{12} - 696 b^{10} + 6504 b^8 + 840 b^6 \\
& - 254 b^4) a^{20} + (-3584 b^{23} + 12288 b^{21} + 36352 b^{19} - 27136 b^{17} - 102400 b^{15} - 41984 b^{13} \\
& + 30208 b^{11} + 15872 b^9 - 1536 b^7) a^{19} + (-4 b^{30} - 48 b^{28} - 148 b^{26} + 2072 b^{24} + 8340 b^{22} \\
& - 71968 b^{20} - 152232 b^{18} - 313656 b^{16} - 369576 b^{14} - 145440 b^{12} - 20844 b^{10} + 8472 b^8 \\
& + 2156 b^6 - 304 b^4 - 4 b^2) a^{18} + (512 b^{25} - 9216 b^{23} - 512 b^{21} + 60416 b^{19} + 50176 b^{17} \\
& - 61952 b^{15} - 82432 b^{13} - 14336 b^{11} + 7680 b^9 + 512 b^7) a^{17} + (512 b^{25} - 7680 b^{23} \\
& - 14336 b^{21} + 25088 b^{19} + 67072 b^{17} + 28672 b^{15} - 25600 b^{13} - 22016 b^{11} - 3072 b^9 \\
& + 512 b^7) a^{15} + (-4 b^{30} - 48 b^{28} + 620 b^{26} - 4072 b^{24} - 17260 b^{22} - 136480 b^{20} \\
& - 262056 b^{18} - 229432 b^{16} - 166056 b^{14} - 78624 b^{12} - 16748 b^{10} - 1512 b^8 - 148 b^6 \\
& - 48 b^4 - 4 b^2) a^{14} + (-1536 b^{25} - 4608 b^{23} - 5632 b^{21} + 1024 b^{19} + 19456 b^{17} \\
& + 30208 b^{15} + 14848 b^{13} - 2048 b^{11} - 2560 b^9) a^{13} + (2 b^{28} + 72 b^{26} - 5272 b^{24} \\
& - 13240 b^{22} - 54424 b^{20} - 128368 b^{18} - 109596 b^{16} - 50800 b^{14} - 18072 b^{12} - 2744 b^{10} \\
& + 360 b^8 + 72 b^6 + 2 b^4) a^{12} + (-2560 b^{25} - 6144 b^{23} - 2048 b^{21} + 4608 b^{19} + 5632 b^{17} \\
& + 7168 b^{15} + 7168 b^{13} + 2560 b^{11}) a^{11} + (-1024 b^{25} - 4608 b^{23} - 7680 b^{21} - 5120 b^{19} \\
& + 1536 b^{15} + 512 b^{13}) a^9 + (-1024 b^{23} - 4096 b^{21} - 6144 b^{19} - 4096 b^{17} - 1024 b^{15}) a^7
\end{aligned}$$

#Plot the curve $r1=0$ in region A2.

$r := a - b :$

with(plots) :

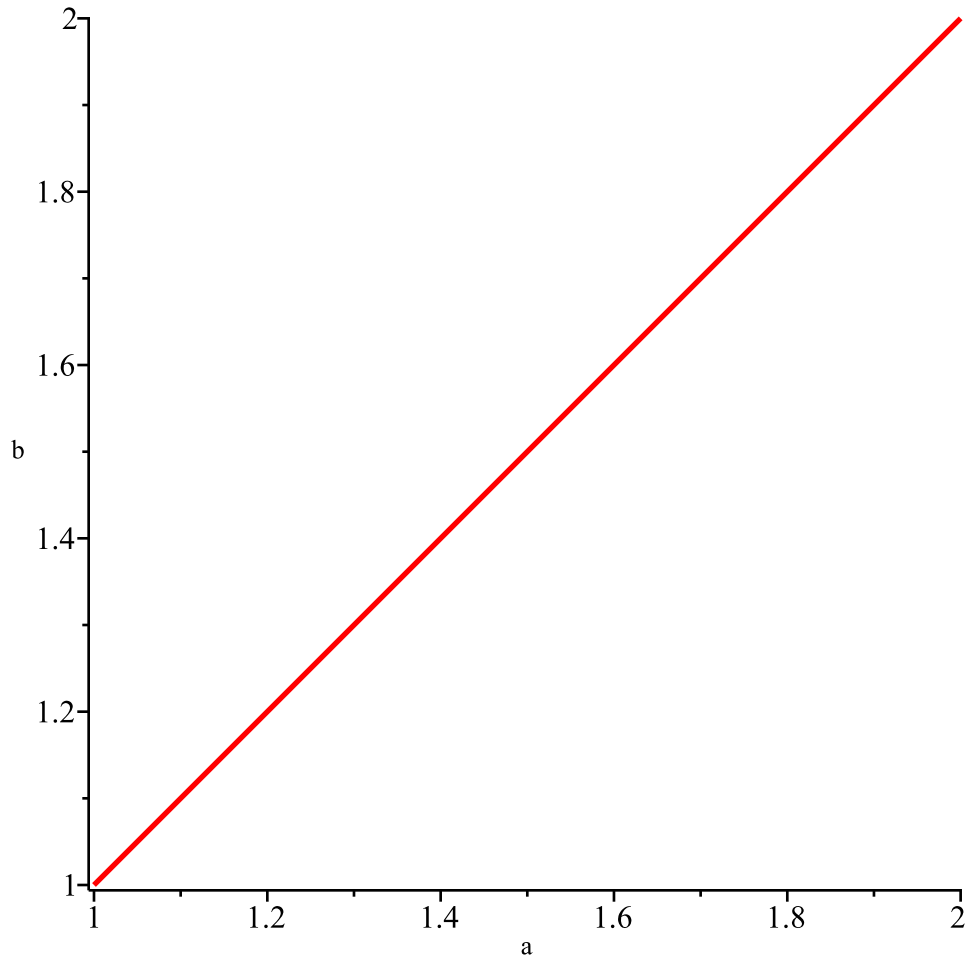
implicitplot([$r1=0, r=0$], $a=1 \dots 2, b=1 \dots (2)$,
scaling=constrained,
gridrefine=2,

```

thickness = 2,
color = [blue, red, black],
labels = ["a", "b"],
title = "Graphic for r1=0 in A2");

```

Graphic for r1=0 in A2



#Apply the Möbius transformation on the rectangle $A2 = [1, 2) \times [1, 2)$ constructing $r1$

$$\text{moebius} := \left\{ \begin{aligned} a &= \frac{1 + 2c}{1 + c}, \\ b &= \frac{1 + (2)d}{1 + d} \end{aligned} \right\} :$$

$r1 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, r1))), c) :$

$$degree(r\hat{l}, c);$$

(4)

$$seq(evalf(assign(cat('c', i), coeff(r1, c, i))), i = 0 .. 32);$$

```
seq(degree(eval(cat('c', i)), d), i = 0 .. 32);
```

[illegible]
$$seq(sign(evalf(coeff(c32, d, i))), i = 32 .. 0, -1);$$
$$\begin{aligned} & -1, \quad (6) \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$
$$seq(sign(evalf(coeff(c31, d, i))), i = 32 .. 0, -1);$$
$$\begin{aligned} &-1, \\ &-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1 \end{aligned} \quad (7)$$
$$seq(sign(evalf(coeff(c30, d, i))), i = 32 .. 0, -1);$$
$$\begin{aligned} & -1, \quad \textbf{(8)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$
$$seq(sign(evalf(coeff(c29, d, i))), i = 32 .. 0, -1);$$
$$\begin{aligned} & -1, \\ & \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned} \tag{9}$$
$$seq(sign(evalf(coeff(c28, d, i))), i = 32 .. 0, -1);$$
$$-1, \quad (10)$$
$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$
$$seq(sign(evalf(coeff(c27, d, i))), i = 32 .. 0, -1);$$
$$\begin{aligned} & -1, \textbf{(11)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$
$$seq(sign(evalf(coeff(c26, d, i))), i = 32 .. 0, -1);$$
$$\begin{aligned} & -1, \quad \textbf{(12)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

[illegible]

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$$\begin{aligned} & seq(sign(evalf(coeff(c23,d,i))), i=32..0,-1); \\ & -1, \textbf{(15)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

$$\begin{aligned} &seq(sign(evalf(coeff(c22,d,i))), i=32..0,-1); \\ &-1, \textbf{(16)} \\ &\quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c2l, d, i))), i=32 .. 0, -1); \\ & -1, \textbf{(17)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c20,d,i))), i=32 .. 0, -1); \\ & \textcolor{blue}{-1,-1}, \quad \textbf{(18)} \\ & \textcolor{blue}{-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1} \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c19,d,i))), i=32..0,-1); \\ & -1, \textbf{(19)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(cI8, d, i))), i=32 .. 0, -1); \\ & \textcolor{blue}{-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,} \quad \textbf{(20)} \\ & \textcolor{blue}{-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1} \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c17,d,i))), i=32..0,-1); \\ & -1, \textbf{(21)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(cI6, d, i))), i=32 .. 0, -1); \\ & -1, \textbf{(22)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c15, d, i))), i = 32 .. 0, -1); \\ & -1, \textbf{(23)} \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c14, d, i))), i=32..0, -1); \\ & -1, \quad (24) \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

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$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

restart

#Define r1

$$\begin{aligned} r1 := & \left(b^{16} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \right. \\ & + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} + 2 b^4 (b^{24} \\ & + 36 b^{22} - 2636 b^{20} - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 \\ & - 1372 b^6 + 180 b^4 + 36 b^2 + 1) a^{12} - 1024 b^{15} (b^2 + 1)^4 a^7 - 4 b^2 (b^{28} + 12 b^{26} + 37 b^{24} \\ & - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} + 36360 b^{10} + 5211 b^8 \\ & - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 2396 b^{18} - 25164 b^{16} \\ & - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 + 420 b^2 - 127) a^{20} \\ & + 8 b^6 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 + 2369 b^6 - 699 b^4 \right. \\ & + b^2 + 1 \left. \right) a^{22} - 9 b^8 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 - 528 b^4 + \frac{4168}{9} b^2 \right. \\ & + 1 \left. \right) a^{24} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} \\ & + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} \\ & - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\ & - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} \\ & - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\ & + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\ & + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\ & + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\ & - 4 b^{14}) a^2 + a^{32} b^{16} - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 + 1)^4 (b^2 - 2) a^{23} - 1536 \left(b^8 - b^6 + \frac{5}{3} b^4 \right. \\ & - \frac{16}{3} b^2 + \frac{5}{3} \left. \right) b^9 (b^2 + 1)^4 a^{13} - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 - 1 \right) b^{11} (b^2 + 1)^4 a^{11} \\ & - 1024 b^{13} (b^2 + 1)^5 \left(b^2 - \frac{1}{2} \right) a^9 + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 \end{aligned}$$

